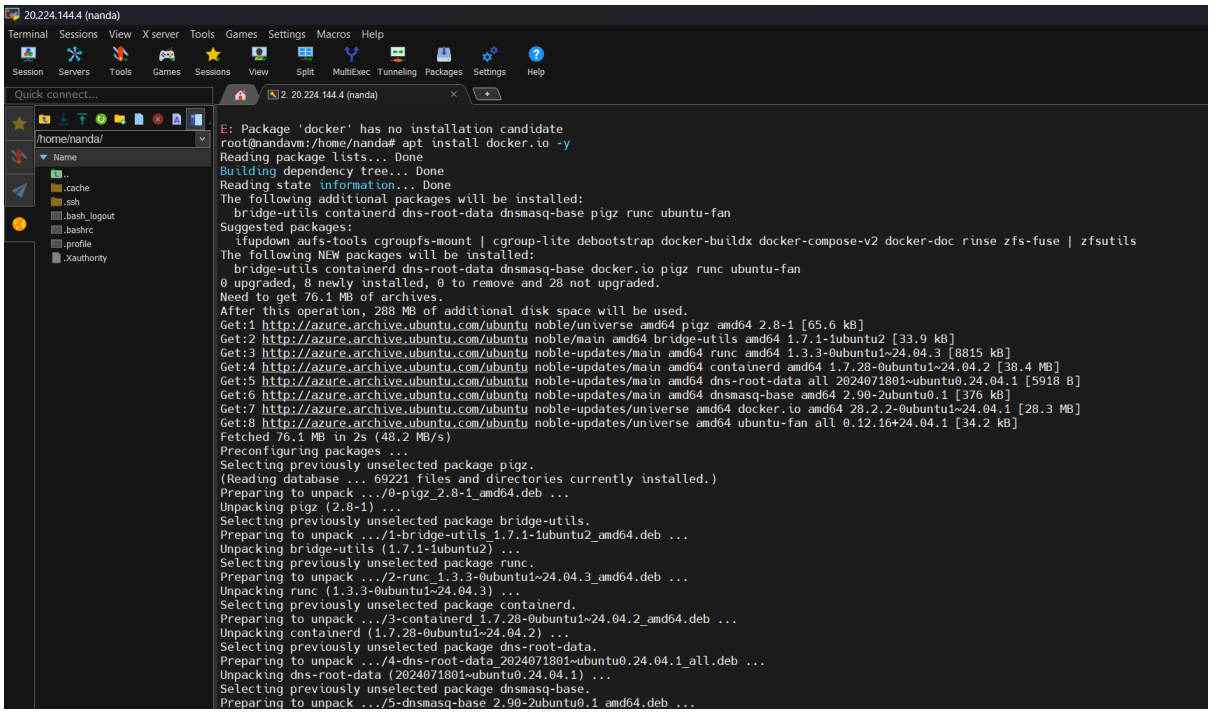
DOCKER

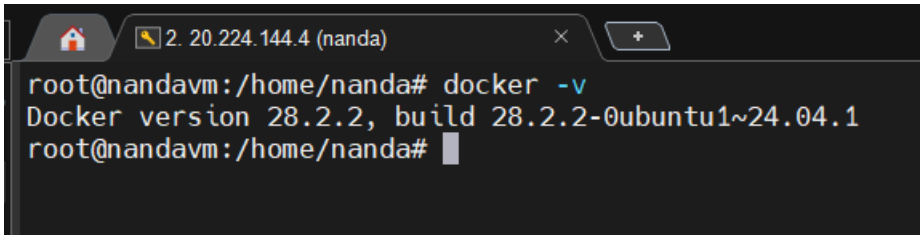
✓ DOCKER is a Containerization Tool.

➢ Installing Docker

- apt install docker.io -y



➢ Version of Docker.



```
Quick connect...
2. 20.224.144.4 (nanda)
/home/nanda/
Name
..
.cache
.ssh
.bash_logout
.bashrc
.profile
.Xauthority

root@nandavm:/home/nanda# docker -v
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
root@nandavm:/home/nanda# docker images
REPOSITORY    TAG        IMAGE ID    CREATED    SIZE
root@nandavm:/home/nanda# docker ps
CONTAINER ID    IMAGE      COMMAND    CREATED    STATUS     P
root@nandavm:/home/nanda# ps -ef | grep dockerd
root         4495       1  0 06:51 ?        00:00:00 /usr
containerd=/run/containerd/containerd.sock
root         8809    2077  0 07:02 pts/1    00:00:00 grep
root@nandavm:/home/nanda# ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 06:45 ?        00:00:05 /sbi
root           2       0  0 06:45 ?        00:00:00 [kth
root           3       2  0 06:45 ?        00:00:00 [poo
root           4       2  0 06:45 ?        00:00:00 [kwo
root           5       2  0 06:45 ?        00:00:00 [kwo
root           6       2  0 06:45 ?        00:00:00 [kwo
root           7       2  0 06:45 ?        00:00:00 [kwo
root           8       2  0 06:45 ?        00:00:00 [kwo
root          10       2  0 06:45 ?        00:00:00 [kwo
root          11       2  0 06:45 ?        00:00:00 [kwo
root          12       2  0 06:45 ?        00:00:00 [kwo
root          13       2  0 06:45 ?        00:00:00 [kwo
root          14       2  0 06:45 ?        00:00:00 [kso
root          15       2  0 06:45 ?        00:00:00 [rcu
root          16       2  0 06:45 ?        00:00:00 [rcu
root          17       2  0 06:45 ?        00:00:00 [rcu
root          18       2  0 06:45 ?        00:00:00 [mig
root          19       2  0 06:45 ?        00:00:00 [idl
root          20       2  0 06:45 ?        00:00:00 [cpu
root          21       2  0 06:45 ?        00:00:00 [cpu
root          22       2  0 06:45 ?        00:00:00 [idl
root          23       2  0 06:45 ?        00:00:00 [mig
root          24       2  0 06:45 ?        00:00:00 [kso
```

- Docker ps it shows the running instances.

```
2. 20.224.144.4 (nanda)
root@nandavm:/home/nanda# docker ps
CONTAINER ID    IMAGE      COMMAND    CREATED    STATUS     PORTS      NAMES
root@nandavm:/home/nanda#
```

- Docker ps -a it shows the created ones.

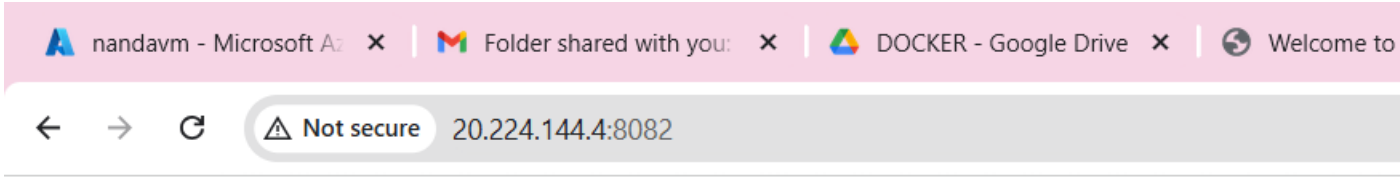
```
2. 20.224.144.4 (nanda)
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID    IMAGE      COMMAND    CREATED    STATUS     PORTS      NAMES
root@nandavm:/home/nanda#
```

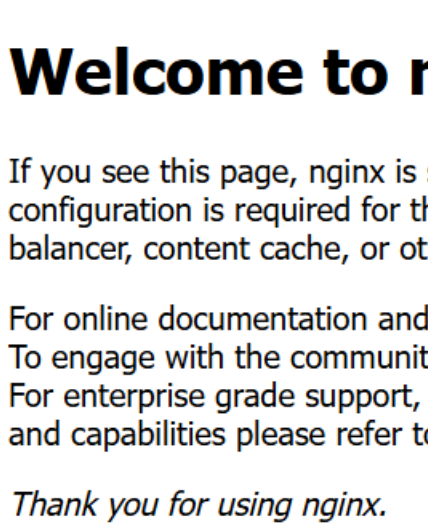
- Installed 3 nginx softwares using docker.

```
2. 20.224.144.4 (nanda)

root@nandavm:/home/nanda# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@nandavm:/home/nanda# docker run -d -p8080:80 nginx
0372a82136dee4fe913eeb234ff92e71c6fe55fb81221ce60e5183ca6f7fb1d1
root@nandavm:/home/nanda# docker run -d -p8081:80 nginx
170fb55f9c6e7b342fdf42a5804212a95ada0aade56d9442dd39586c446eebc3
root@nandavm:/home/nanda# docker run -d -p8082:80 nginx
9d978d20c7ca164308f280d096b91079b4ffc3af404b5ca3879505cfcfea5778
```

```
2. 20.224.144.4 (nanda)

root@nandavm:/home/nanda# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS
 PORTS                                       NAMES
320c8c2fefac   httpd     "httpd-foreground"       4 minutes ago   Up 4 minutes
 0.0.0.0:8083->80/tcp, [::]:8083->80/tcp   mystifying_moser
9d978d20c7ca   nginx     "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes
 0.0.0.0:8082->80/tcp, [::]:8082->80/tcp   naughty_goldberg
170fb55f9c6e   nginx     "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes
 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp   great_pascal
0372a82136de   nginx     "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes
 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   quirky_margulis
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS         PORTS
7bb916f27170   httpd     "httpd-foreground"       3 minutes ago   Created
78923f36cfba   httpd     "httpd-foreground"       4 minutes ago   Created
320c8c2fefac   httpd     "httpd-foreground"       4 minutes ago   Up 4 minutes   0.0.0
9d978d20c7ca   nginx     "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes   0.0.0
170fb55f9c6e   nginx     "/docker-entrypoint.…"   5 minutes ago   Up 5 minutes   0.0.0
0372a82136de   nginx     "/docker-entrypoint.…"   6 minutes ago   Up 6 minutes   0.0.0
root@nandavm:/home/nanda#
```

- 3 nginx Softwares are installed.(8080,8081,8082)





- To Stop those, use command docker stop id

```
root@nandavm:/home/nanda# docker stop 320c8c2fefac
320c8c2fefac
root@nandavm:/home/nanda# docker stop 9d978d20c7ca
9d978d20c7ca
root@nandavm:/home/nanda# docker stop 170fb55f9c6e
170fb55f9c6e
root@nandavm:/home/nanda# docker stop 0372a82136de
0372a82136de
root@nandavm:/home/nanda# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS
7bb916f27170   httpd     "httpd-foreground"       16 minutes ago   Created
78923f36cfba   httpd     "httpd-foreground"       17 minutes ago   Created
320c8c2fefac   httpd     "httpd-foreground"       18 minutes ago   Exited (0) About a m
9d978d20c7ca   nginx     "/docker-entrypoint.…"   19 minutes ago   Exited (0) About a m
170fb55f9c6e   nginx     "/docker-entrypoint.…"   19 minutes ago   Exited (0) 46 second
0372a82136de   nginx     "/docker-entrypoint.…"   19 minutes ago   Exited (0) 25 second
root@nandavm:/home/nanda#
```

- To delete those, we use docker rmi nginx

- To know any corrupted images, we have to use the command,

docker images -f dangling=true

```
root@nandavm:/home/nanda# docker images
REPOSITORY   TAG       IMAGE ID       CREATED       SIZE
nginx        latest    1a1e63136420   6 days ago    161MB
httpd        latest    95e97c51ad04   6 days ago    117MB
ubuntu       latest    f794f40ddfff   3 weeks ago   78.1MB
root@nandavm:/home/nanda# docker images -f dangling=true
REPOSITORY   TAG       IMAGE ID   CREATED   SIZE
root@nandavm:/home/nanda#
```

- To remove the dargling images,

  docker image prune -y

```
root@nandavm:/home/nanda# docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N]
```

- To remove all images without at least one container associated to them.

  docker image prune -a

```
Quick connect...
2. 20.224.144.4 (nanda)
/home/nanda/
Name
..
.cache
.ssh
.bash_logout
.bashrc
.profile
.Xauthority

root@nandavm:/home/nanda# docker image prune
WARNING! This will remove all dangling images.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B
root@nandavm:/home/nanda# docker image prune -a
WARNING! This will remove all images without at lea
Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: nginx:latest
untagged: nginx@sha256:dec7a90bd0973b076832dc56933
deleted: sha256:1a1e6313642049664ed59099f3ca37a672
deleted: sha256:78468d93cdb85e379561b2bd790094d11f
deleted: sha256:d8909058a77df3864878b175665aa9d239
deleted: sha256:522ced56ca8be99e612c3e2ebd1a09a250
deleted: sha256:285017e72e61d2562bdbd56b6c289c60fc
deleted: sha256:7c4c916aa492138ccc0d1d274a8798508f
deleted: sha256:fa716e18f6f407d86f7c5dc9b6dbd9ce8f
untagged: httpd:latest
untagged: httpd@sha256:331548c5249bdeced0f048bc2fb
deleted: sha256:95e97c51ad0493de94e31f59bb02af16af
deleted: sha256:a0105fbdb32e3628e55269f616dfbcc978
deleted: sha256:bcd5f1aa90930e4ff4ef5a43318862b2cd
deleted: sha256:6c685ec41c12b5325f0f9cb85574d2e68e
deleted: sha256:68d116713b24cb0e0792a70864b86d95c7
deleted: sha256:9930d8a74f22cbb14e8d244e45d8afd438
deleted: sha256:188c9b34dfbe022075d01fc4f5a3054129
untagged: ubuntu:latest
untagged: ubuntu@sha256:186072bba1b2f436cbb91ef256
deleted: sha256:f794f40ddfff5af8ef1b39ee29eab3b540
deleted: sha256:f2a7f072635332d307212e318e07284948

Total reclaimed space: 277.2MB
root@nandavm:/home/nanda#
```

- BY USING IMAGES HOW WE CREATE CONTAINERS

✓ To Create a container we use command,

docker create –name web01 nginx

```
2. 20.224.144.4 (nanda)
rt a terminal game
root@nandavm:/home/nanda# docker create --name web01 nginx
a69188b71849175b8abda1e988d45350b8f059c23ce7987b97cf28c70ce027ed
```

```
root@nandavm:/home/nanda# docker create --name web01 nginx
a69188b71849175b8abda1e988d45350b8f059c23ce7987b97cf28c70ce027ed
root@nandavm:/home/nanda# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS    POR
a69188b71849   nginx     "/docker-entrypoint.…"   59 seconds ago   Created
root@nandavm:/home/nanda#
```

✓ We have to start the container

```
root@nandavm:/home/nanda# docker create --name web01 nginx
a69188b71849175b8abda1e988d45350b8f059c23ce7987b97cf28c70ce027ed
root@nandavm:/home/nanda# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS    POR
a69188b71849   nginx     "/docker-entrypoint.…"   59 seconds ago   Created
root@nandavm:/home/nanda# docker start web01
web01
root@nandavm:/home/nanda# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED         STATUS
a69188b71849   nginx     "/docker-entrypoint.…"   2 minutes ago   Up 9 seconds
root@nandavm:/home/nanda# docker stop web01
web01
root@nandavm:/home/nanda# docker rm web01
web01
root@nandavm:/home/nanda# docker images
REPOSITORY   TAG       IMAGE ID       CREATED      SIZE
nginx        latest    1a1e63136420   6 days ago   161MB
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@nandavm:/home/nanda#
```

```
root@nandavm:/home/nanda# docker pull httpd
Using default tag: latest
latest: Pulling from library/httpd
ec781dee3f47: Already exists
09625190bc81: Pull complete
4f4fb700ef54: Pull complete
a1897530540b: Pull complete
753dca1c3a38: Pull complete
5184cf4f524b: Pull complete
Digest: sha256:331548c5249bdeced0f048bc2fb8c6b6427d2ec6508bed9c1fec6c57d0b27a60
Status: Downloaded newer image for httpd:latest
docker.io/library/httpd:latest
root@nandavm:/home/nanda# docker create --name web02 httpd
ba2e7febc2214bbd2c70b4496559a0ffd97461b83bae4d8dc1d8e4384c98fb7b
root@nandavm:/home/nanda# docker start web02
web02
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND              CREATED          STATUS         PO
ba2e7febc221   httpd     "httpd-foreground"   19 seconds ago   Up 6 seconds   80
root@nandavm:/home/nanda# docker stop web02
web02
root@nandavm:/home/nanda# docker rm web02
web02
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@nandavm:/home/nanda#
```

When we pull and create ubuntu it will works but when we try to start we got an error

```
root@nandavm:/home/nanda# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
817807f3c64e: Pull complete
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
root@nandavm:/home/nanda# docker create --name web03 ubuntu
5393d111e4666c10bef234b469bc5dfb65c3b3c46354219d4ba7ecab3835314a
root@nandavm:/home/nanda# docker start web03
web03
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS
5393d111e466   ubuntu    "/bin/bash"   57 seconds ago   Exited (0) 40 seconds ago
root@nandavm:/home/nanda# docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
root@nandavm:/home/nanda# docker ps -a
CONTAINER ID   IMAGE     COMMAND       CREATED              STATUS
5393d111e466   ubuntu    "/bin/bash"   About a minute ago   Exited (0) 57 seconds
root@nandavm:/home/nanda#
```

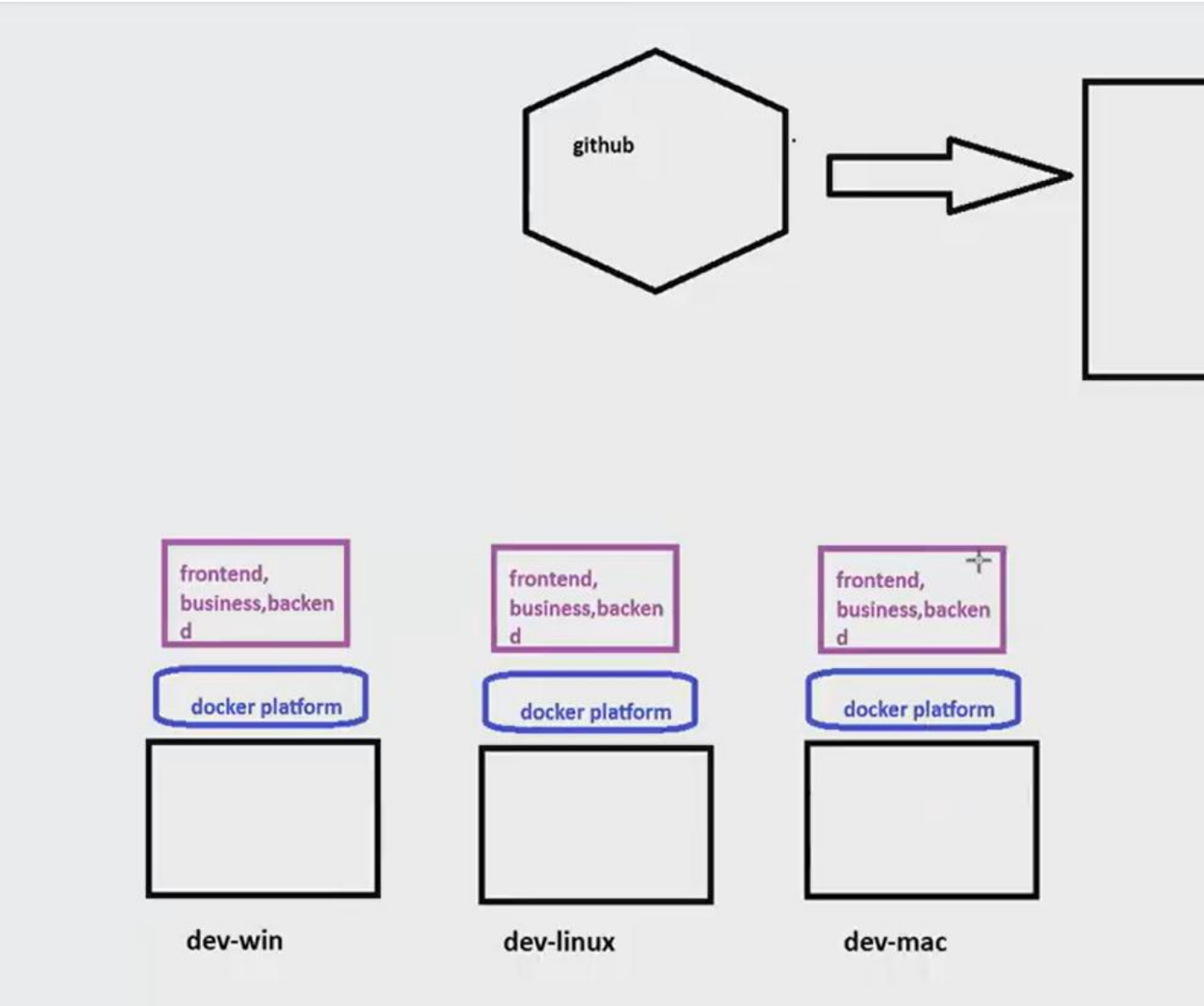
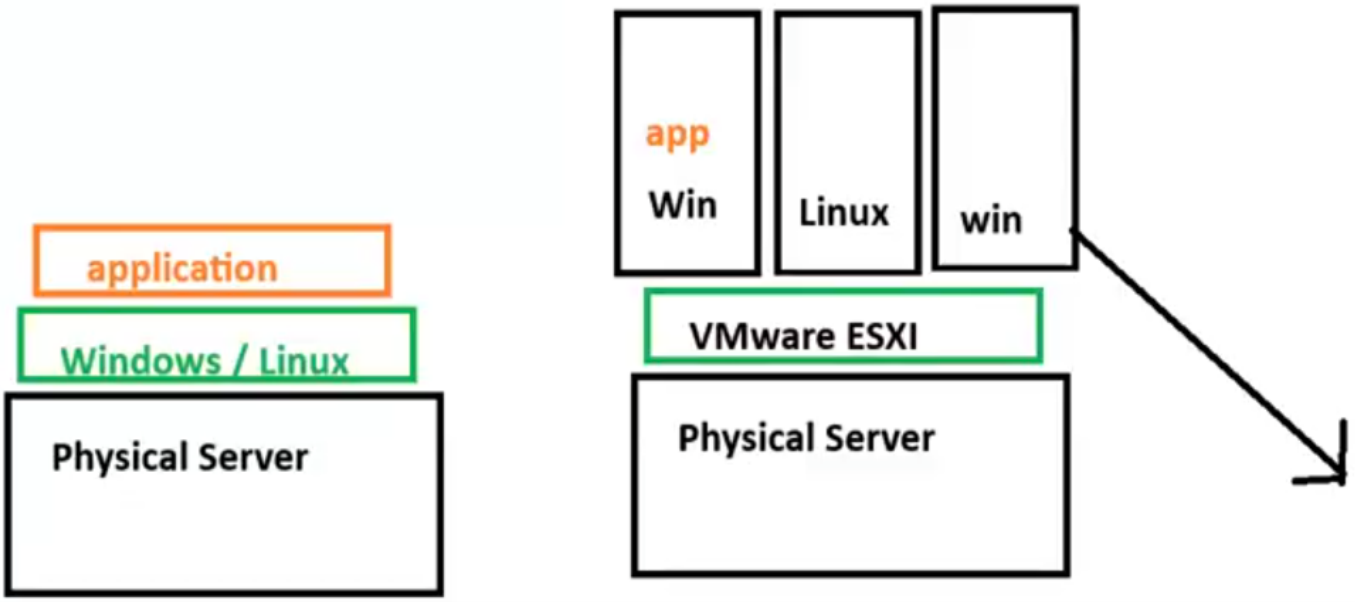
✓ When we try to start the ubuntu we get Exited.

- ✓ Exited occurs because when we create a container using ubuntu image, the cmd -> **/bin/bash** it will works as a background so we cant continue the container.
- ✓ Ubuntu works on background.
- ✓ To resolve this we have to use interactive terminal while creating the container only.
- ✓ Use -it to interactive terminal
- ✓ **Command – docker run -dit --name web01 ubuntu**

```
2. 51.124.247.54 (nanda)
root@nanda-vm:/home/nanda# docker run -dit --name web01 ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
817807f3c64e: Pull complete
Digest: sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c
Status: Downloaded newer image for ubuntu:latest
5ee4dfe825deb17284dda9e26ac7867d4c56e74a9d426c0f7a858fbcd85b3561
root@nanda-vm:/home/nanda#
```

❖ NOTE

- ✓ If we want to delete the image first stop & delete the container, then stop & delete the image.
- ✓ Delete image as rmi
- ✓ Delete container as rm


github
frontend, business,backen d
docker platform
dev-win
frontend, business,backen d
docker platform
dev-linux
frontend, business,backen d
docker platform
dev-mac

Virtualization:
virtualize OS
layer